quarto_practice

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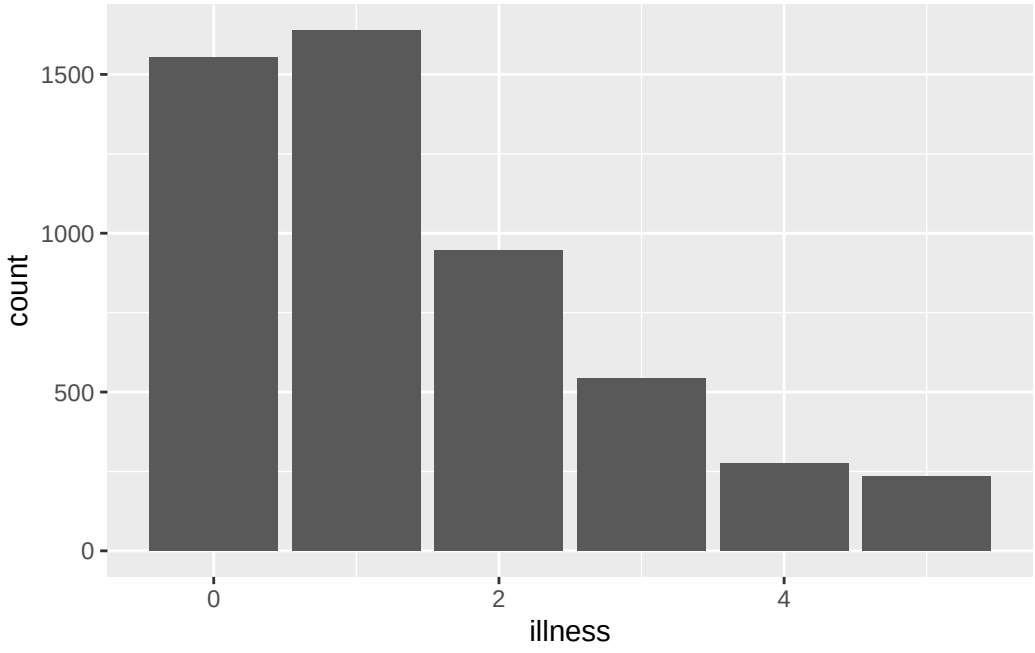
This is my abstract. I am following Alexander (2023), to use R Core Team (2021)

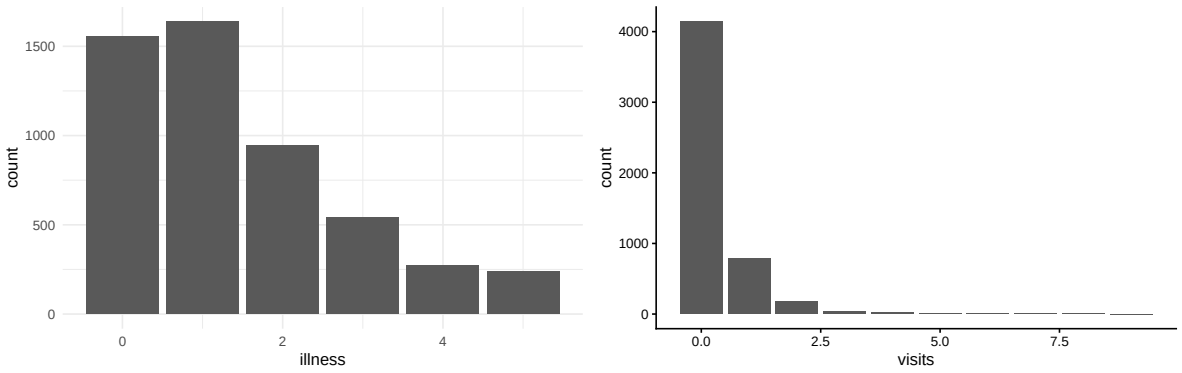
```
#install.packages("AER")
#install.packages("great_tables", repos = "https://cloud.r-project.org")
library(tidyverse)
library(AER)
library(tinytable)
data("DoctorVisits", package = "AER")

DoctorVisits |>
  ggplot(aes(x = illness)) +
  geom_histogram(stat = "count")
```



```r
DoctorVisits |>
  ggplot(aes(x = illness)) +
  geom_histogram(stat = "count") +
  theme_minimal()

DoctorVisits |>
  ggplot(aes(x = visits)) +
  geom_histogram(stat = "count") +
  theme_classic()
```



(a) Illnesses

(b) Visits to the doctor

Figure 1: Two variants of graphs

$$
\begin{aligned}
y_i|\pi_i &\sim \text{Bern}(\pi_i) \\
\text{logit}(\pi_i) &= \beta_0 + \alpha^{\text{gender}}_{g[i]} + \alpha^{\text{age}}_{a[i]} + \alpha^{\text{state}}_{s[i]} + \alpha^{\text{edu}}_{e[i]} \\
\beta_0 &\sim \text{Normal}(0, 2.5) \\
\alpha^{\text{gender}}_g &\sim \text{Normal}(0, 2.5) \text{ for } g = 1, 2 \\
\alpha^{\text{age}}_a &\sim \text{Normal}\left(0, \sigma^2_{\text{age}}\right) \text{ for } a = 1, 2, \dots, A \\
\alpha^{\text{state}}_s &\sim \text{Normal}\left(0, \sigma^2_{\text{state}}\right) \text{ for } s = 1, 2, \dots, S \\
\alpha^{\text{edu}}_e &\sim \text{Normal}\left(0, \sigma^2_{\text{edu}}\right) \text{ for } e = 1, 2, \dots, E \\
\sigma_{\text{gender}} &\sim \text{Exponential}(1) \\
\sigma_{\text{state}} &\sim \text{Exponential}(1) \\
\sigma_{\text{edu}} &\sim \text{Exponential}(1)
\end{aligned}
$$

Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. https://tellingstorieswithdata.com.

R Core Team. 2021. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.